

FOR LITTLE EXPLORERS

Leaves Changing Color

Why do leaves change color?

READ ALOUD

Swish. Sway. Rustle. (Sway your hands like leaves.) *Big Wonderer and Little Wonderer stand beneath a tree. Yesterday's green branches have patches of gold.*

All summer, a green helper inside the leaves caught sunlight to help make food for the tree. Its green color hid other colors.

ASK **Can you spot a leaf with more than one color?**

Now autumn days are shorter. The green helper breaks down, and the leaves stop making more. **Hello, yellow! Hello, orange!** *Those colors were there all along.* (Open your hands for a surprise.)

Some leaves make red in autumn, too. ASK **Which color would you point to first?**

A breeze shakes the branch. One leaf lets go and twirls down. On the ground, fallen leaves slowly become part of the soil.

ASK **How could your hand move like a falling leaf?**

Big Wonderer holds out a hand. Little Wonderer takes it. In spring, this tree will grow fresh green leaves.

Inside: a leaf story · color science · first-then-last · spot the difference · a maze · counting to 20 · shapes and colors · rhymes and letter sounds · tracing and snipping · a coloring page · and a fold-up picture book.

A tree changes its coat



Green leaves catch the light

In summer, leaves use sunlight to help make food for the tree. A green helper called chlorophyll catches the light and makes the leaves look green.

ASK **Point to a leaf catching the sunshine.**



The green begins to fade

As autumn days grow shorter, many trees get ready for winter. The green helper breaks down, and the leaves stop replacing it. Other colors begin to show.

ASK **Follow the leaf's edge with your finger. Can you find green and yellow?**



Hello, hidden colors

Yellow and orange colors were inside the leaves all along. With less green hiding them, we can see them more easily.

ASK **Can you find a yellow leaf beside an orange one?**

TRY IT WITH A GROWN-UP First, look at a low leafy branch with your grown-up. Then, point to all the colors you can see without picking the leaves.

Where do the colors come from?

A leaf holds colors inside it. As the green fades, some colors show through. Some leaves make a new color, too.



Yellow was hiding

Yellow coloring is already inside many green leaves. When the green fades, the yellow becomes easier to see.



Orange was hiding

Orange coloring is already inside many leaves, too. Less green can reveal a warm orange glow.



Some leaves make red

Some trees make red coloring in their leaves during autumn. Different trees can show different mixes of colors.



A blanket of leaves

Many trees let their autumn leaves fall. On the ground, the leaves slowly break down and become part of the soil.

ASK **Point: can you find yellow, orange, or red?**

(Some trees stay green. Every tree has its own seasonal story.)

Round and round. *This tree rests through winter with bare branches. In spring, new green leaves unfold. Summer brings a leafy canopy, and autumn brings another color change.*

EXPERIMENT WITH A GROWN-UP First, choose a low leafy branch to watch with your grown-up. Then, visit it together over several days and compare its colors. Do all the leaves change together?

Which leaf goes with which tree?

Three leafy trees: one green, one yellow, one red. Each has a matching leaf.

1 Use your finger to join each leaf to the tree of the same color.



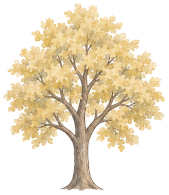
a green leaf



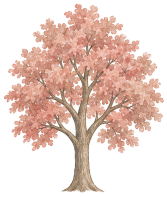
a yellow leaf



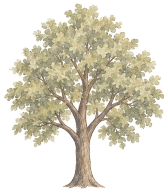
a red leaf



yellow leaves



red leaves



green leaves

WHY? These pictures match by color. A real tree can wear several leaf colors at once.

2 Shapes and colors beneath the tree

Point to each shape, then say its color with your grown-up.



yellow circle

The round sun looks like a circle.



red triangle

A pointy leaf tip looks like a triangle.



brown square

A flat path stone is a square.



green oval

A rounded leaf looks like an oval.

ASK Can you find something oval in your room?

What happened first?

Grown-up: point to each picture and tell it in a few words. Tell the story together using First, Then, and Last. Little explorer circles 1, 2 or 3 under each picture.

1 From summer green to autumn gold



1 2 3



1 2 3



1 2 3

First, green leaves catch sunlight. Then, green fades. Last, gold shows.

2 Leaves settle on the ground



1 2 3



1 2 3



1 2 3

First, golden leaves hang. Then, leaves fall. Last, leaves cover the ground.

TALK ABOUT IT What might this tree look like when spring comes?

Spot 5 differences

The two pictures look the same, but 5 things changed. Point to each one, then circle it in the bottom picture. Color a dot for each one you find.

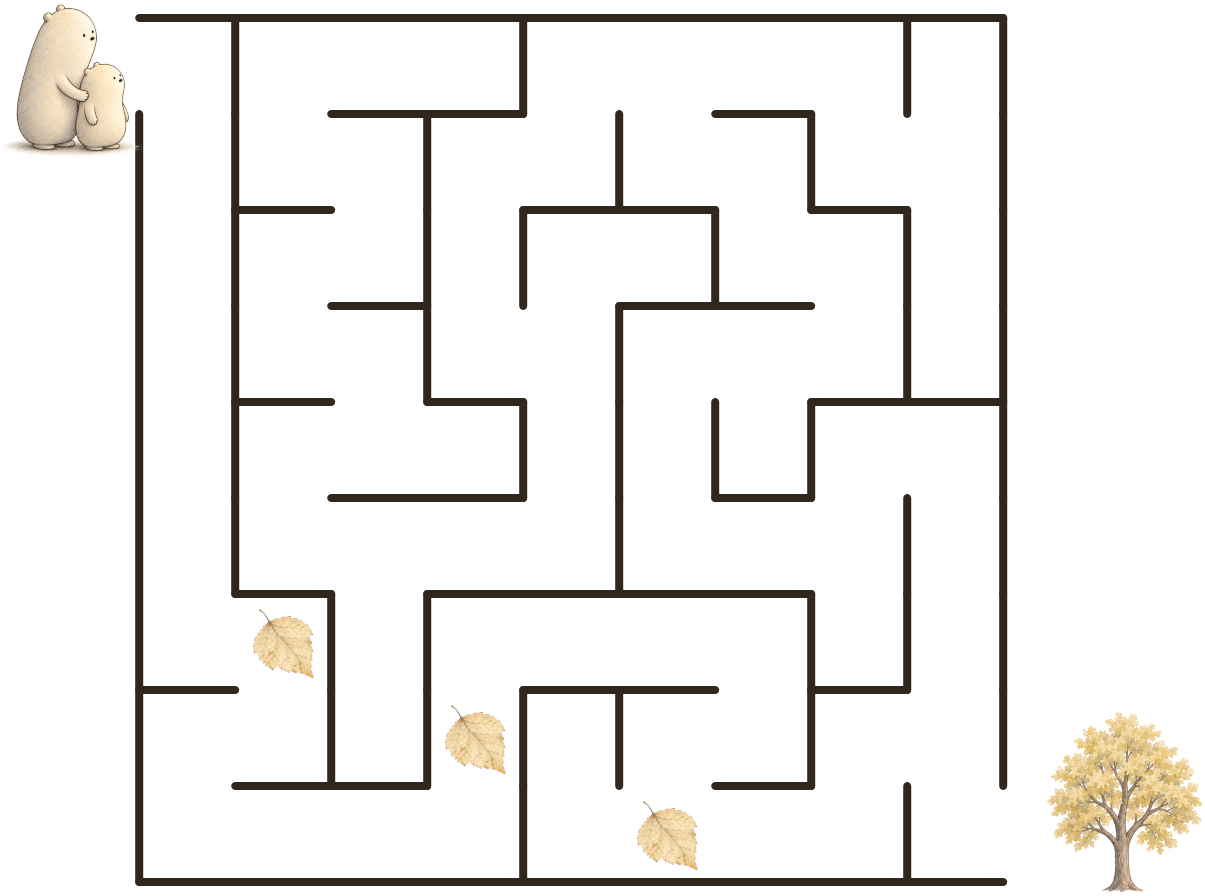


I found: ○ ○ ○ ○ ○

Follow the autumn path

Big and Little Wonderer are walking to the golden tree. Help them find the way and spot the three leaf pictures along the path.

First, trace the way with a finger.
Then, touch each leaf as you pass it.
Last, follow the way again with your finger.

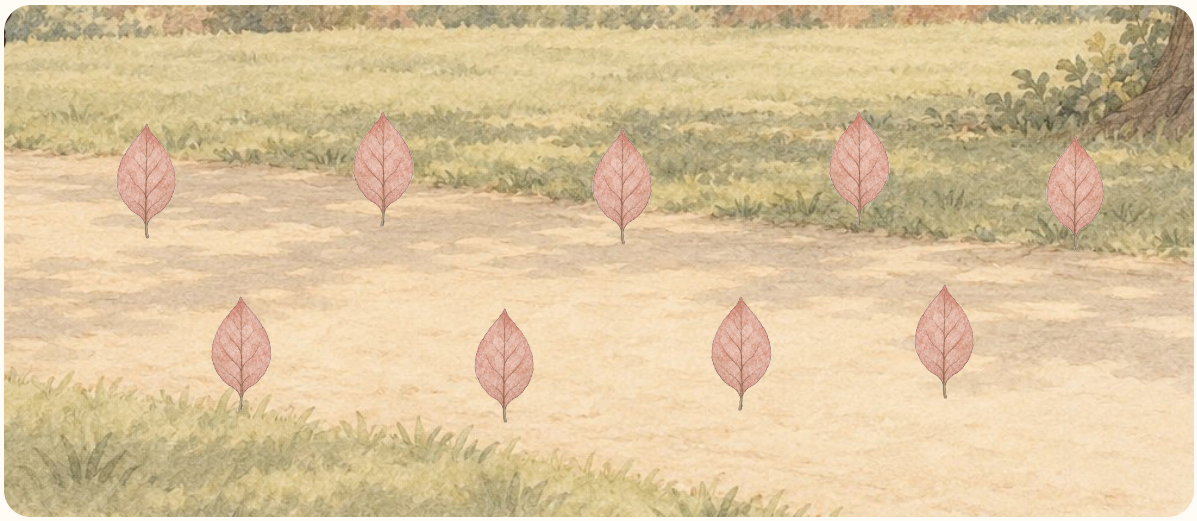


Leaves spotted:

WHY DO LEAVES FALL? Many trees seal off the place where each leaf joins its twig. The join weakens, and the leaf can fall. This helps the tree save water through winter.

Autumn leaf count

1 Count the red leaves. Circle how many.

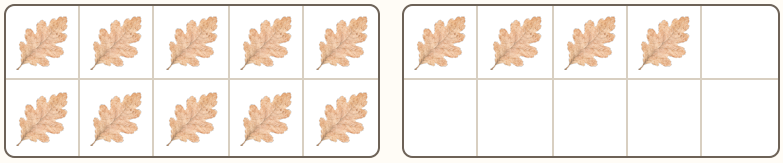


Touch each leaf once as you count. Put a crayon dot on each leaf you touch.

8 9 10 12

2 Count the orange leaves. Circle the number.

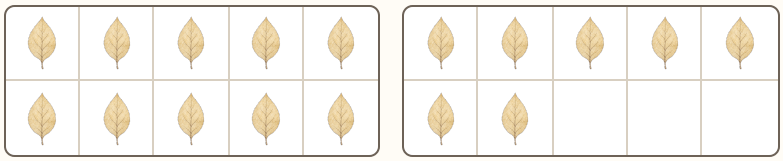
One full frame is 10. Then count on: 11, 12…



13 14 16

3 Count the yellow leaves. Circle the number.

10 and some more.



15 17 20

4 Find the number 12. Circle it.

7 12 15 20 10

5 Point to the group with MORE.



Autumn picture stories

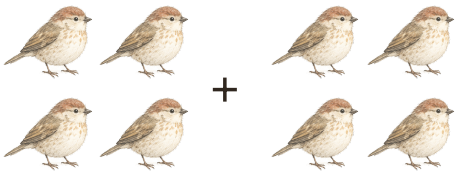
A stretch, just for fun. Grown-up reads each little story. Count the first group, then count on with the second group. Little explorer circles the answer.

3 yellow leaves rest on the path. 2 more flutter down. How many yellow leaves now?



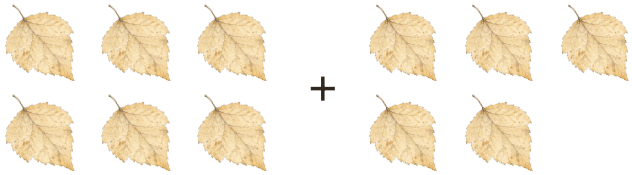
3 + 2 = (4) (5) (6)

4 birds sit on one autumn branch. 4 birds sit on another. How many birds?



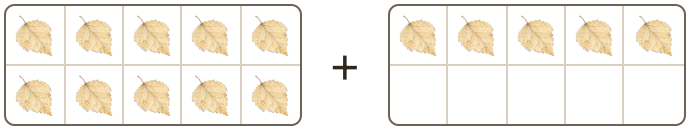
4 + 4 = (7) (8) (9)

6 leaves lie beneath the tree. Then 5 more fall. How many leaves?



6 + 5 = (10) (11) (13)

10 yellow leaf pictures fill the frame. 5 more leaf pictures join them. How many leaves?



10 + 5 = (14) (15) (16)

Sounds beneath the leaves

B b

This is the letter B, b. It says /b/, like the start of bug. Point to each B, b together.

1 **Which pictures start with /b/, like *bug*? Circle them.**


bug


sun


bee


leaf


bed


log

2 **Rhyme time! Grown-up says the three words. Point to the two that rhyme.**




cat hat sun




bug leaf mug




log bee tree

3 **Read it together. Point to each word as you say it.**


A bug sits on a log. 


The cat sits by a log. 

4 **Just for fun: touch each letter, then say the word fast.**


b u g


c a t

Trace and snip

Count the pictures together. Trace the first number with a finger, then trace the dotted numbers with a crayon. Start at the dot.



1 1 1 1



2 2 2 2



3 3 3 3



4 4 4 4



5 5 5 5

Snip along the leafy breeze

With a grown-up helping, snip along the wavy line from the leaf to the tree. Safety scissors, and a grown-up holds the paper.





Color the autumn leaves

First, color the big oval leaf yellow.

Then, color the other leaves any autumn colors you like.

Last, circle the big oval leaf.

Try to stay inside the lines. Big and Little Wonderer can be any color you like.



Leaves Changing Color

FOR LITTLE EXPLORERS

Why do leaves change color?

1

Green leaves catch the summer sunlight.

2

Autumn comes. The green begins to fade.

3

Hidden colors begin to show.

4

Yellow was there all along.

5

Orange was hiding in the leaves, too.

6

Some leaves make red in autumn.

7

Leaves flutter down. Spring brings new green.

8

How to fold the picture book

1. Print the picture sheet on one page, landscape, at 100% (no scaling).
2. Fold it in half the long way, then open it. Fold it in half the short way, then in half again; open it flat.
3. **Grown-up only:** fold it in half the short way and cut along the middle fold from the folded edge to the center crease.
4. Open it, fold it the long way again, and push the ends together so the slit opens into a little book. Page 1 goes on the front.

Read it together: one picture, a few words on each page. Ask "what do you see?" before you turn the page.